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Optimization of DDPG positioning and navigation technology for warehouse logistics robots

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ABSTRACT

To address the core limitations of Deep Deterministic Policy Gradient (DDPG) algorithms in warehouse logistics robot navigation—specifically, inadequate temporal memory, inaccurate Q-value evaluation, and low training sample efficiency—this paper proposes a novel multidimensional optimization scheme. The main novelties include: (1) Integrating Long Short-Term Memory (LSTM) networks into the Actor-Critic framework to enhance historical state utilization and decision continuity; (2) Adopting an improved dueling network architecture to decouple the Q-value into state value and action advantage, thereby refining critic evaluation accuracy; and (3) Designing a dynamic dual-pool experience replay mechanism that prioritizes high-value samples based on TD-error and reward dimensions to optimize training efficiency. These three components form a synergistic optimization system. Experimental results demonstrate that the optimized algorithm significantly outperforms the original DDPG. It achieved a trajectory coherence index of 0.9 (a 38 % improvement) and an obstacle avoidance success rate of 83 %. The dueling network reduced the Q-value mean squared error to 2×10^{-3}, while the dynamic dual-pool mechanism increased key sample utilization to 89.3 % and accelerated convergence speed by 38 %. In practical tests, the algorithm exhibited a dynamic obstacle avoidance response time of 210 ms, a path optimality index of 0.95, and a multi-robot task delay of only 5.2 s. The proposed multidimensional optimization strategy significantly enhances the adaptability, reliability, and navigation efficiency of warehouse logistics robots in complex dynamic environments, providing a robust solution for intelligent warehousing systems. Additionally, we discuss safety/ethical requirements and deployment scalability (compute and communication constraints) for large robot fleets.

1. Introduction

With the rapid development of the logistics industry, warehouse automation has become the key to improving logistics efficiency [1]. As the core equipment of modern warehousing systems, warehouse robots have achieved an intelligent operation mode of "goods to people" through autonomous navigation and precise positioning technology, greatly improving the efficiency of warehousing operations [2,3]. Among the various technologies of warehouse robots, path planning, as the core technology of autonomous motion, directly determines the operational efficiency and safety of the robot [4,5]. Beyond efficiency and safety, practical deployment in real warehouses introduces critical constraints. In human-robot mixed environments, ensuring safe, transparent, and reliable operations is paramount. Furthermore, scaling to coordinate dozens or hundreds of robots necessitates careful consideration of computational and communication costs, which are key factors for industrial adoption. The development of path planning technology has gone through an evolutionary process from static environment to dynamic environment [6]. Early path planning methods were mainly based on geometric search algorithms. Jiang et al. proposed an optimal path planning model based on the Dijkstra algorithm to address the complex and unpredictable behavior of crowds in indoor emergency evacuation. The results showed that the detection accuracy of a single multi-box detector was 78.2 %, and the accuracy and precision of deep simple online real-time tracking were 71.2 % and 78.5 %, respectively. The model can adjust the path direction in real time, adapt to dynamic changes in the crowd, and provide an efficient intelligent solution for indoor evacuation [7]. Hu et al. proposed a multi-stage path planning technique to address the challenges of difficult tasks and high skill requirements in the automatic docking operation of underactuated surface vehicles. Experimental verification showed that this method had good path planning and tracking performance, and could effectively assist in the automatic berthing of ships and the development of intelligent ships [8]. Liu et al. proposed an asynchronous genetic algorithm to address

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the problems of long planning time, slow convergence, and unstable solutions in multi-agent path planning using traditional genetic algorithms. The simulation results showed that although this algorithm did not provide the optimal solution, it significantly outperformed other algorithms in terms of execution speed and solution stability, effectively addressing the shortcomings of traditional genetic algorithms [9]. Pu et al. proposed an improved ant colony algorithm to solve multi-objective path planning problems for single and multiple robots. The experimental results showed that in single robot path planning, the improved ant colony algorithm outperformed traditional ant colony algorithms in terms of path length, convergence speed, and running time [10].

With the breakthrough of deep learning technology, path planning methods based on deep reinforcement learning have demonstrated stronger environmental adaptability and real-time decision-making ability, providing a new technological path for intelligent navigation in complex dynamic environments [11]. Cui et al. proposed an Improved Q-Learning Ant Colony Optimization (IQLACO) algorithm to solve the problems of low search efficiency and susceptibility to local optima in traditional ant colony optimization algorithms in path planning. The experimental results showed that IQLACO outperformed other algorithms in terms of optimal path length, convergence speed, and turning time, demonstrating higher stability and optimization ability [12]. Li and Chai proposed a factory logistics path planning model based on grid environment and ant colony optimization algorithm to meet the flexibility, efficiency, and punctuality requirements of warehouse logistics robots in the Industry 4.0 era. The results showed that the improved algorithm had a fast convergence speed, with the shortest path length reduced from 26.46 to 24.38 and a maximum reduction of 18.44 %. Moreover, the path had good continuity after smoothing with B-spline curves [13]. Lyu proposed an improved Deep Deterministic Policy Gradient (DDPG) method based on deep reinforcement learning to enhance the autonomous path planning capability of warehouse logistics robots in unfamiliar environments. The research results provide a new strategy for efficient autonomous navigation of warehouse logistics robots in complex environments [14]. LSTM-based intelligent prediction has been shown to improve control robustness by forecasting future system responses and compensating delay effects [15]. Transformer-based bidirectional prediction further demonstrates the effectiveness of deep sequence models for adaptive decision-making under complex non-stationary excitations [16].

In summary, the existing DDPG algorithm faces three core problems in path planning for warehouse logistics robots: firstly, the network structure lacks temporal memory ability, resulting in a lack of continuity in robot action decisions. Secondly, the evaluation accuracy of the Q-value by the evaluation network is insufficient, which affects the optimization effect of the action strategy. Finally, the traditional experience replay mechanism has low sampling efficiency and insufficient utilization of important samples. To address these issues, a multidimensional optimization scheme for DDPG is proposed: introducing Long Short Term Memory (LSTM) network to construct a spatiotemporal feature extraction module, enhancing state memory capability. Adopting an improved dueling network architecture, the Q-value evaluation accuracy is improved through collaborative optimization of state values and action advantage functions. A dynamic adaptive sampling mechanism based on dual experience pools is designed to intelligently adjust the sample collection strategy according to the training progress. These optimization measures are aimed at improving the path planning performance of the algorithm in complex warehouse environments. The main novelty of this paper is a unified multidimensional optimization of DDPG for warehouse navigation that simultaneously enhances temporal decision continuity (LSTM), Q-value estimation stability (improved dueling critic), and sampling efficiency (dynamic dual-pool replay). The main contributions are: (1) LSTM-based spatiotemporal feature extraction for improved temporal memory; (2) improved dueling critic for more accurate Q-value evaluation; (3) dynamic adaptive dual-pool sampling for efficient training and better utilization of high-value experiences.

2. Methods and materials

2.1. DDPG architecture optimization based on LSTM and duel network

With the increasing demand for adaptability to dynamic environments in the intelligent logistics industry, the shortcomings of traditional path planning methods are gradually becoming apparent. Deep reinforcement learning techniques, especially the DDPG algorithm, have become a research hotspot due to their advantages in processing continuous action spaces [17]. However, this algorithm has three core issues: temporal memory loss, Q-value evaluation bias, and low exploration efficiency [18]. To address these challenges, research proposes multidimensional optimization strategies. An LSTM module is integrated to construct a spatiotemporal feature extractor, enabling the robot to analyze historical states. The duel network architecture is reconstructed to accurately separate state value and action advantage. A Gaussian and Ornstein-Uhlenbeck Process Noise (OU) mixed noise mechanism is designed to balance exploration breadth and depth. LSTM serves as a key component, utilizing a gating mechanism to intelligently filter temporal information, as shown in Fig. 1.

In Fig. 1, LSTM achieves long-term dependency modeling of temporal data through the transfer of hidden states and memory units between input sequences and time steps. As a typical long sequence decision-making task, warehouse robot path planning requires high temporal processing capability of the algorithm. The traditional DDPG algorithm cannot effectively utilize historical state information and is prone to getting stuck in local optimal solutions under limited perception range constraints [19]. To address this deficiency, research is being conducted on integrating LSTM modules into the DDPG architecture to construct a novel spatiotemporal fusion network. The improved network structure is shown in Fig. 2.

Fig. 2 shows the improved LSTM-DDPG network architecture, which achieves intelligent decision-making through dual network collaboration. The Actor network uses LSTM layers to process temporal features of states and outputs robot control instructions. The Critic network analyzes the state value through LSTM and evaluates the Q-value by combining action inputs. However, traditional Q-value calculation methods are difficult to distinguish between the inherent value of environmental states and the marginal benefits brought by actions, resulting in systematic biases in evaluations. For this purpose, a dueling network architecture is adopted in the Critic network, which decomposes the Q-value into two independent components: state value V (v) and action advantage A (a). The state value reflects the basic quality of the environment, while the action advantage represents the additional benefits brought by a specific action. After normalization, the two are linearly combined to output the final Q value. Firstly, the schematic diagram of the dueling network architecture is shown in Fig. 3.

In Fig. 3, the input state is split through the shared layer to the dominance network and the value network, generating action dominance A (a) and state value V (v), respectively. The two are fused to output an accurate Q-value. This structure enables the storage robot to independently evaluate the environmental state and action contribution, and quickly identify the optimal obstacle avoidance strategy in complex scenarios. Afterwards, the dueling network will be introduced into the Critic network to improve the accuracy of navigation decisions through separate value evaluation. Its structure is shown in Fig. 4.

In Fig. 4, the improved Critic network architecture adopts a dual channel dueling design to process state and action inputs. The state information is extracted by LSTM to extract temporal features and input into the V-network to evaluate the basic value of the environment. The action data is input into the A-network through a fully connected layer to calculate the relative advantage. The two outputs are standardized and fused to generate a Q-value, achieving independent quantification

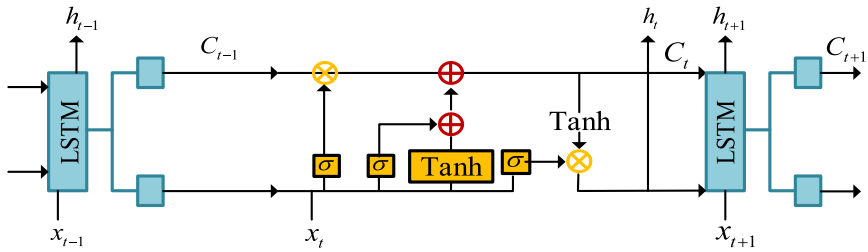


Fig. 1. LSTM structure diagram.

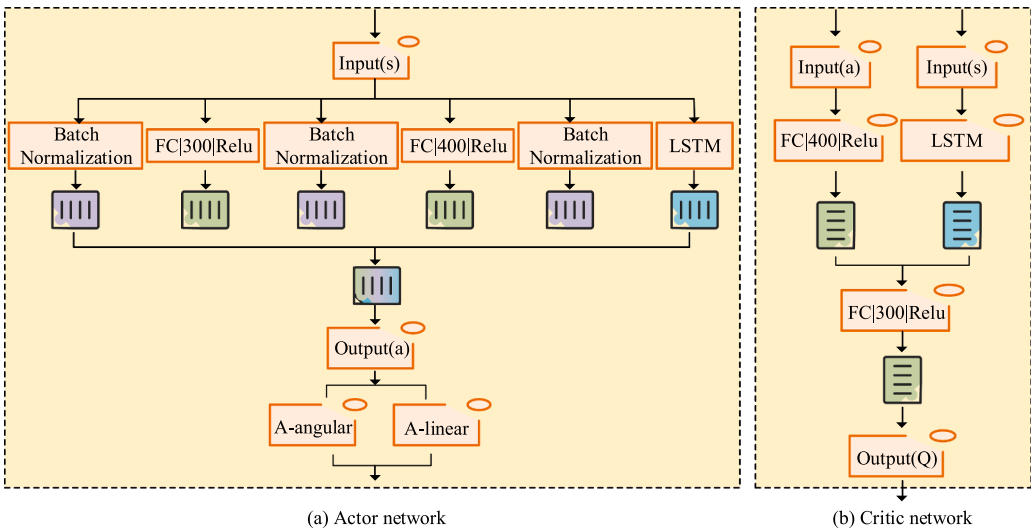


Fig. 2. LSTM-DDPG network.

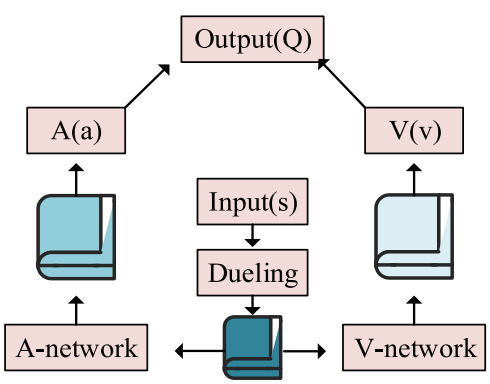


Fig. 3. Dueling network architecture.

of value and advantage. To enhance the generalization ability of the model and prevent overfitting, a series of regularization strategies were introduced during the training process. After the LSTM layers of the Actor and Critic networks, a Dropout layer (with a dropout rate of 0.2) was added, and L2 weight decay (with a coefficient of 1e-4) was applied to all fully connected layer parameters. Gradient clipping (with a threshold of 1.0) was adopted to stabilize the training process of the LSTM. Additionally, an entropy regularization term based on the variance of the action distribution ($Lentropy = \beta \cdot \log(\sigma_a)$) was introduced in the loss function of the Actor network. The coefficient β linearly decreases with the number of training steps to encourage the exploration diversity in the early training stage.

After introducing the dueling network, it was found that the traditional structure still suffers from the problem of mutual interference between state values and action dominance values. The simple addition of the two leads to non unique Q-value evaluation results, which affects training stability. To this end, a new improvement plan is proposed by introducing the average value of the extreme value of the action advantage function as a regulating factor, while maintaining the essence of the function unchanged, to ensure the stability of the ranking of action advantages in different states. The improved Q-value calculation is shown in Eq. (1).

$$\begin{cases} Q = V(v) + A(a) - Avg(A(a)) \\ A(a)_{Avg} = \dfrac{A(a)_{Max} + A(a)_{Min}}{2} \end{cases} \tag{1}$$

In Eq. (1), $A(a)_{Avg}$ represents the extreme mean of the action advantage function. $A(a)_{Max}$ represents the maximum action advantage value in the current state. $A(a)_{Min}$ represents the minimum action advantage value in the current state. Although LSTM and duel networks enhance memory and evaluation capabilities, a single exploration strategy still limits global search efficiency. Therefore, studying the fusion of Gaussian noise and OU noise to construct a hybrid exploration mechanism enhances the path discovery capability in complex environments by balancing random perturbations and directional exploration. The state space of the storage robot obtains environmental feedback through visual sensors, and its detection range is the front [-90 °, 90 °] area. The minimum detection distance is 0.2 m, and any distance less than this is considered a collision. The 10 dimensional environmental data collected by the sensor constitutes the state input, as shown in Eq. (2).

$$angle_{range} = [angle_1, \ldots, angle_{10}] \tag{2}$$

In Eq. (2), n is the effective perception distance range of the robot vision sensor. $angle_n$ is the environmental feature data collected by the

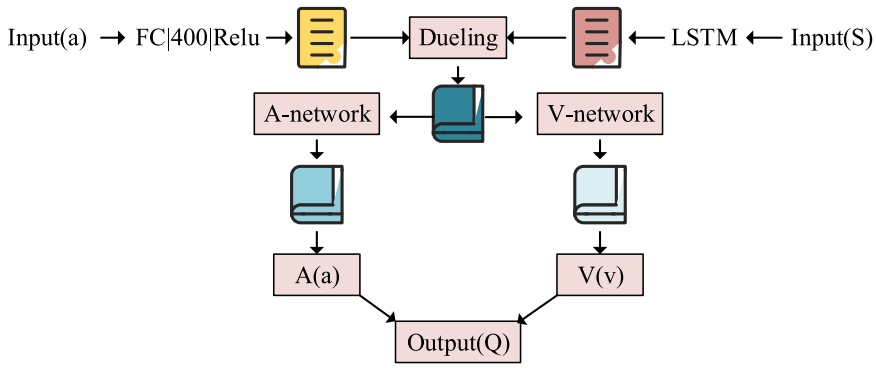


Fig. 4. Critic network structure after introducing duel network.

sensor at the nth detection direction. The path planning of storage robots requires the integration of multi-dimensional states such as obstacle information, historical actions, relative position and attitude angle of target points. To improve learning efficiency, the study used differential normalization method to preprocess various state variables, and the specific transformation relationship is shown in Eq. (3).

$$\begin{cases} rel_{dis} = rel_{dis}/dia_{dis} \\ rel_{angle} = rel_{angle}/360 \\ yaw_{angle} = yaw_{angle}/360 \\ Tar_{angle} = Tar_{angle}/180 \end{cases} \tag{3}$$

In Eq. (3), Tar_{angle} is the target spread angle. rel_{angle} is the relative angle. yaw_{angle} is the yaw angle. rel_{dis} stands for relative distance. dia_{dis} is the diagonal length of the map. The state space of the storage robot is normalized to form the network input state, as shown in Eq. (4).

$$S_m = [v_{m-1}, \varphi_{m-1}, rel_{dis}, rel_{angle}, yaw_{angle}, Tar_{angle}, angle_{range}] \tag{4}$$

In Eq. (4), v is the line speed. φ stands for angular velocity. m stands for Time. Due to the limitations of the simulation environment, the maximum angular velocity of the robot is 0.5 rad/s, and the linear velocity is 0.25 m/s. Its output is shown in Eq. (5).

$$a_m = [v_m, \varphi_m] \tag{5}$$

In Eq. (5), a_t represents the behavior at m time. The DDPG algorithm adopts a deterministic strategy, which leads to insufficient exploration and requires adding noise to enhance learning ability in action output. Traditional methods use Gaussian noise or OU noise. The former generating independent random exploration, while the latter achieves temporal correlation but is prone to local wandering. To optimize exploration efficiency, a mixed noise mechanism is studied and designed, which combines the wide area randomness of Gaussian noise and the directional persistence of OU noise to generate more efficient action outputs through Eq. (6).

$$a_m \rightarrow N_{Gao}[a_m + N_{OU}(d_{a_m}), \mathrm{var}] \tag{6}$$

In Eq. (6), var is Gaussian variance. N_G is Gaussian noise. N_{OU} stands for OU process noise. d_{a_m} stands for motion disturbance.

2.2. Adaptive sampling training optimization based on dynamic dual pool

The optimization of the multidimensional network structure through LSTM and dueling networks has enhanced the temporal memory and Q-value evaluation capabilities of DDPG, thereby improving the path planning performance of warehouse robots. However, the training efficiency is still limited by the uniform random sampling of traditional experience replay mechanisms, which cannot distinguish differences in sample value, resulting in low utilization of high-value data. To address this bottleneck, a dynamic adaptive sampling strategy is proposed. This strategy collaborates deeply with the LSTM-DDPG network, where the LSTM module provides accurate state evaluation through spatiotemporal feature extraction, and the dynamic sampling mechanism ensures efficient learning from these evaluations.

To clearly describe how the LSTM module supports the proposed dynamic sampling strategy, the temporal feature output of the LSTM is denoted as h_t, which is obtained by feeding the recent state sequence $\{s_{t-K+1}, \ldots, s_t\}$into the Actor-Critic LSTM layer. Based on h_t, the Critic estimates $Q(h_t, a_t)$, and the temporal-difference (TD) error is computed as $\delta_t = r_t + \gamma Q(h_{t+1}, \mu(h_{t+1})) - Q(h_t, a_t)$. Thus, the TD-error reflects the "unexpectedness" of the current transition under temporal memory, while the reward reflects the "task success quality". These two dimensions are jointly used to identify high-value experiences, which is more suitable than single-criterion replay for dynamic warehouse navigation.

The experience replay of traditional DDPG only uses single-pool random or priority sampling, with the former ignoring sample differences and the latter prone to edge state instability. Warehousing and logistics samples possess dual value: high TD-error samples contain crucial strategy information, and high-reward samples represent successful trajectories. Balancing these in a single pool is difficult and leads to slow convergence. To this end, a dual experience pool architecture is proposed, as shown in Fig. 5, which intelligently manages samples based on TD-error and reward values, synchronously optimizing the two-dimensional value.

In Fig. 5, the architecture is based on a dual standard of TD-error absolute value and reward value for sample screening, with high-value samples stored in the D1 pool and ordinary samples stored in the D2 pool. Its advantage lies in the dynamic hierarchical management of empirical data, which retains the strategic improvement value of key samples and ensures training stability through an intelligent diversion mechanism. Dual-pool storage rule: Each transition (h_t, a_t, r_t, h_{t+1}) is assigned to the high-value pool D_1 or normal pool D_2 using two thresholds α_δ and α_r: If $|\delta_t| \geq \alpha_\delta$ or $r_t \geq \alpha_r$, the sample is stored in D_1; otherwise it is stored in D_2. To avoid scale sensitivity, both $|\delta_t|$ and r_t are normalized within the sliding window using min-max normalization before thresholding. As shown in Eq. (7).

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{7}$$

Where xrepresents either $|\delta_t|$or $r_t$within the current window, and $x_{\min}$and $x_{\max}$are the minimum and maximum values of xin that window, respectively. The normalized values are denoted as $\delta'$and r'_t.

Due to the dynamic nature of the storage environment, the importance of samples may vary with the training process, and fixed priority standards may lead to sampling bias in the later stages. To this end, an adaptive weighting mechanism is designed and optimized through three stages: dynamic evaluation system, mixed sampling strategy, and stability control.

Firstly, the dynamic evaluation system introduces a sliding window statistical method to track changes in sample value distribution in real time. The classification thresholds $\alpha_\delta$and $\alpha_r$are updated online based on

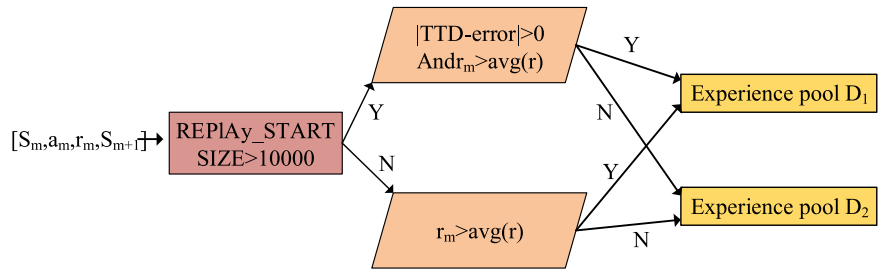


Fig. 5. Dual experience pool architecture.

the mean and standard deviation of the normalized indicator x within a window of size W, As shown in Eq. (8).

$$\tau_t = \mu_t + \beta \cdot \sigma_t$$

$$\tau_t = \frac{1}{W}\sum_{i=t-W+1}^{t} v_i + \alpha \cdot \sqrt{\frac{1}{W}\sum_{i=t-W+1}^{t}(v_i - \mu_t)^2} \tag{8}$$

In Eq. (8), μ_t is the mean value of samples within the window. τ_t is the adaptive classification threshold. β is a sensitivity coefficient. α is the sensitivity coefficient. v_i is a sample value indicator. t is the current number of training steps. i is the time step number of the sample within the sliding window. On the basis of dynamic threshold calculation, to further improve the adaptability of the algorithm to different training stages, a self-adjusting mechanism of introducing window size is studied. Its calculation is shown in Eq. (9).

$$W_t = clip(W_{base} \cdot (1 + \frac{\log(1+t)}{\log(1+T)}), W_{\min}, W_{\max}) \tag{9}$$

In Eq. (9), W_{base} is the reference window. $W_{\min}$ and $W_{\max}$ are the upper and lower bounds of the window. T is the total number of training steps. Secondly, the hybrid sampling strategy constructs an intelligent sampling mechanism that adapts to the training phase by dynamically balancing the advantages of priority sampling and uniform sampling. This strategy adopts high weight priority sampling in the early stage of training, gradually transitioning to uniform sampling dominance through the exponential decay law of Eq. (10).

$$\beta_t = \beta_{init} \cdot \gamma^{t/T}, \gamma = 0.9999 \tag{10}$$

In Eq. (10), β_{init} is the weight coefficient with a value of 0.9. β_t represents the sampling weight coefficients of the algorithm during the t_{th} step of training. γ is the attenuation factor. This design enables the algorithm to quickly lock in high-value samples in the early stages and expand state coverage through uniform sampling in the later stages, effectively resolving the contradiction between convergence speed and generalization ability of traditional methods.

Mixed sampling ratio: At each training step, a mini-batch of size B is constructed by sampling $B_1 = [w_t B]$ transitions from $B_2 = B - B_1$ transitions from D_2. This design enables fast early learning from informative trajectories and gradually increases diversity for generalization. (3) Stability control rationale. Because D_1 contains many high-TD-error samples, directly over-sampling them may cause Q-value spikes and unstable critic updates. Therefore, the proposed stability constraint in Eq.(10) clips extreme Q-values within a dynamic confidence interval to suppress outliers while retaining high-value learning signals. Finally, a stability control mechanism is introduced in the training of the Critic network, which effectively solves the training fluctuation problem caused by high priority samples through dynamic value constraints. This mechanism uses sliding window statistics to monitor the distribution of Q-values in real time, and limits the influence of extreme values through the truncation function of Eq. (11).

$$Q_{target} = clip(Q(s,a), \mu_Q - \lambda_t \sigma_Q, \mu_Q + \lambda_t \sigma_Q) \tag{11}$$

In Eq. (11), μ_Q represents the mean value of Q. σ_Q is the standard deviation of Q value. λ_t is the dynamic constraint coefficient, which decays linearly according to Eq. (12).

$$\lambda_t = 3 - 2 \cdot \frac{t}{T} \tag{12}$$

Finally, the overall framework for studying optimization schemes is shown in Fig. 6.

As shown in Fig. 6, the study elucidates the multidimensional optimization scheme of DDPG navigation technology for warehouse logistics robots. In terms of network architecture, the model integrates LSTM to enhance temporal memory, reconstructs the duel network to separate value and advantage, and designs a hybrid noise mechanism to optimize exploration efficiency. At the training level, it innovatively adopts a dual experience pool dynamic sampling strategy, establishes a dual-dimensional evaluation system for TD-error and reward values, and improves training efficiency through adaptive sampling and stability control. The LSTM module provides the environmental cognitive foundation for dynamic sampling, while intelligent sampling accelerates network optimization. Together, they form a comprehensive algorithmic framework, offering a systematic solution for robot navigation in complex warehouse environments.

Deployment Architecture Note: In a practical system, the learned policy's action outputs are processed by a safety layer that enforces hard constraints on velocity, minimum obstacle distance, and emergency stops. For auditability, key decision metrics like state value (V), advantage (A), and TD-error are logged, leveraging the inherent structures of the dueling network and the dual-pool sampling mechanism.

Overall, the proposed dynamic dual-pool mechanism follows three steps: (i) compute TD-error δ_t using LSTM-temporal features h_t, normalize $|\delta_t|$ and r_t, and store transitions into D_1 or D_2 based on the dual-threshold rule; (ii) update thresholds α_δ, α_r and window size W_t online via Eqs. (8) and (9); (iii) sample mixed batches via Eq. (9) and stabilize critic training via Eqs. (11) and (12). Together, they form a comprehensive algorithmic framework, offering a systematic solution for robot navigation in complex warehouse environments.

3. Results

3.1. Cost analysis of calculation and inference

To assess the practical deployment cost, we profile the computational overhead of the optimized DDPG. The inference latency, resource utilization, and model complexity are summarized in Table 1. All metrics are measured on a standard server with an NVIDIA RTX 3090 GPU and an Intel Xeon Gold 6226 R CPU.

3.2. Algorithm performance verification experiment

To verify the impact of LSTM on navigation performance, the experiment was conducted in a dynamic warehouse environment built using Gazebo. The environment size ranged from a randomly generated 150 m× 150 m to 200 m× 200 m layout, and 100 trajectories with

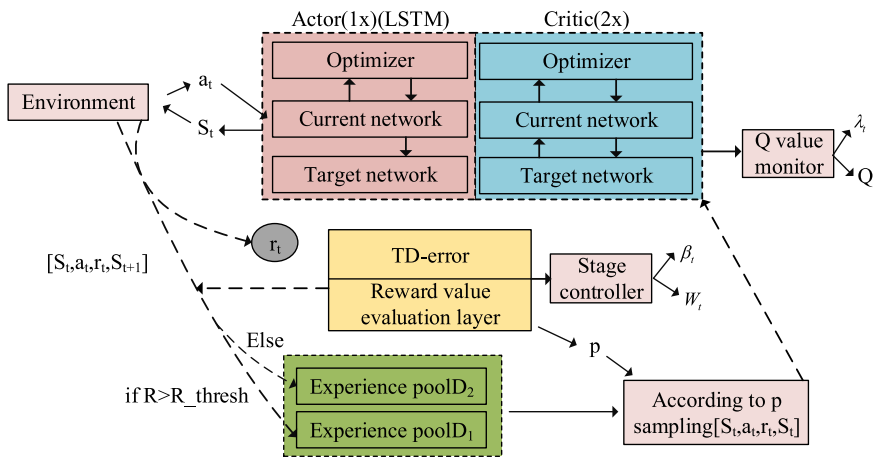


Fig. 6. Framework of multidimensional optimization strategy.

Table 1
Computational Cost and Model Complexity Profiling.

Metric	Original DDPG	Optimized DDPG (LSTM-Dueling)
Inference Latency (ms/step)	8.2 ± 1.1	18.5 ± 2.1
GPU Memory Footprint (MB)	1245	2180
CPU Utilization (%)	15.3 ± 2.5	28.7 ± 3.8
Trainable Parameters (Million)	0.45	2.1
Training Time (hours)	12.5	22.8

dynamically changing speed and moving obstacles were set up. The robot was equipped with a 10-dimensional laser radar and IMU for perception. Four methods, namely the original DDPG, LSTM-DDPG, GRU-DDPG, and Transformer-DDPG, were compared in the experiment. The evaluation was conducted based on trajectory consistency index, historical state utilization rate, obstacle avoidance success rate, and path repetition rate. Eighty navigation tasks were executed in 12 different warehouse layouts, and a collaborative scenario with 50 robots was introduced to test scalability. The data sampling interval was 10 ms, and a 72-hour long-term stability test was carried out to record the performance decay rate and abnormal recovery time to comprehensively evaluate the adaptability and robustness of the algorithm in complex dynamic environments. The experimental results are shown in Fig. 7.

As shown in Fig. 7, the experimental results indicate that LSTM-DDPG outperforms other methods in multiple indicators. In Fig. 7(a), its trajectory consistency index remains within the range of 0.85–0.90. Fig. 7(b) shows that its utilization rate of historical states reaches 62 %–68 %. In Fig. 7(c), its dynamic obstacle avoidance success rate is stable at 82 %–86 %. Fig. 7(d) indicates that its path repetition rate is only 13 %–19 %. Figs. 7(e) and 7(f) show that when the number of robots increases, the obstacle avoidance success rate of LSTM-DDPG decreases more slowly and the path repetition rate is also lower. In Fig. 7(g), its performance attenuation rate remains within the range of 2.5 %–4.5 % per 24 h. Fig. 7(h) shows that its abnormal recovery time gradually increases from approximately 4.5 min to approximately 6.5 min. Overall, LSTM-DDPG performs optimally in terms of trajectory smoothness, information utilization, obstacle avoidance success rate, decision stability, and long-term robustness, verifying the effectiveness of LSTM sequence modeling in improving the navigation performance of warehouse logistics robots. To conduct a thorough analysis of how the LSTM module enhances decision continuity, the study plotted a heat map of the attention weights for historical states during the robot navigation process (see Fig. 8). This graph visually demonstrates the changes in the LSTM network's focus on different time-step historical information when processing sequence states.

Fig. 8 Heatmap of historical state attention weights from the LSTM module during a typical navigation task, showing increased focus on recent states at critical decision points. The results show that when approaching critical decision points such as dynamic obstacles or path forks, LSTM significantly increases the attention weight given to recent historical states (with darker colors), which validates its ability to effectively capture temporal and spatial dependencies and make smooth decisions.

To verify the impact of the dueling network architecture on the accuracy of Q-value evaluation, testing was conducted in a 3-layer stereoscopic shelf warehouse environment constructed by PyBullet. Comparing the original Critic network, Duel network, Double Deep Q-Network (Double DQN), and Dueling Double Deep Q-Network (Dueling Double DQN), four methods were used to evaluate the accuracy of Q-value Mean Squared Error (QMSE), validate the decoupling effect of the network through Advantage Separation Degree (ASD), evaluate the stability of Value Estimation Variance (VEV) response, and evaluate the actual utility through Policy Improvement Rate (PIR). The experiment constructed a test set of 1000 standard state-action pairs, performed forward propagation after freezing the network parameters, and used the Monte Carlo method to obtain the baseline Q-values. The experimental results are shown in Fig. 9.

Fig. 9(a) shows that the QMSE index significantly decreased from 8×10^{-3} in the original Critic network to 2×10^{-3} in the Dueling Double DQN, with a decrease of 75 %, while the ASD index increased from 20 % to 90 %. In Fig. 9(b), the VEV value decreased by 83 % from 6×10^{-2} to 1×10^{-2}, and the PIR index increased from 15 % to 40 %, indicating that network improvement significantly improves the stability of value estimation. The results showed that Dueling Double DQN performed the best in all four core indicators, with QMSE and VEV reaching the lowest 2×10^{-3} and 1×10^{-2}, respectively, while maintaining a high decoupling advantage of 90 % and a strategy improvement rate of 40 %, which is overall superior to other architectures.

To verify the improvement effect of dynamic dual experience pool on training efficiency, it was tested in the logistics sorting scenario constructed by ROS-MoveIt, including conveyor belt system and random package obstacles. Comparing the four methods of uniform playback, priority playback, dynamic dual pool sampling, and mixed layered sampling, the effectiveness of sample selection was measured by the utilization rate of key samples, the training efficiency was evaluated by the required sample size for convergence, the training stability coefficient reflected the smoothness of the learning process, and the empirical playback time recorded the computational cost. The experiment monitors the sampling distribution of each episode, fixes the initial network parameters, tracks the convergence characteristics of the loss curve, and

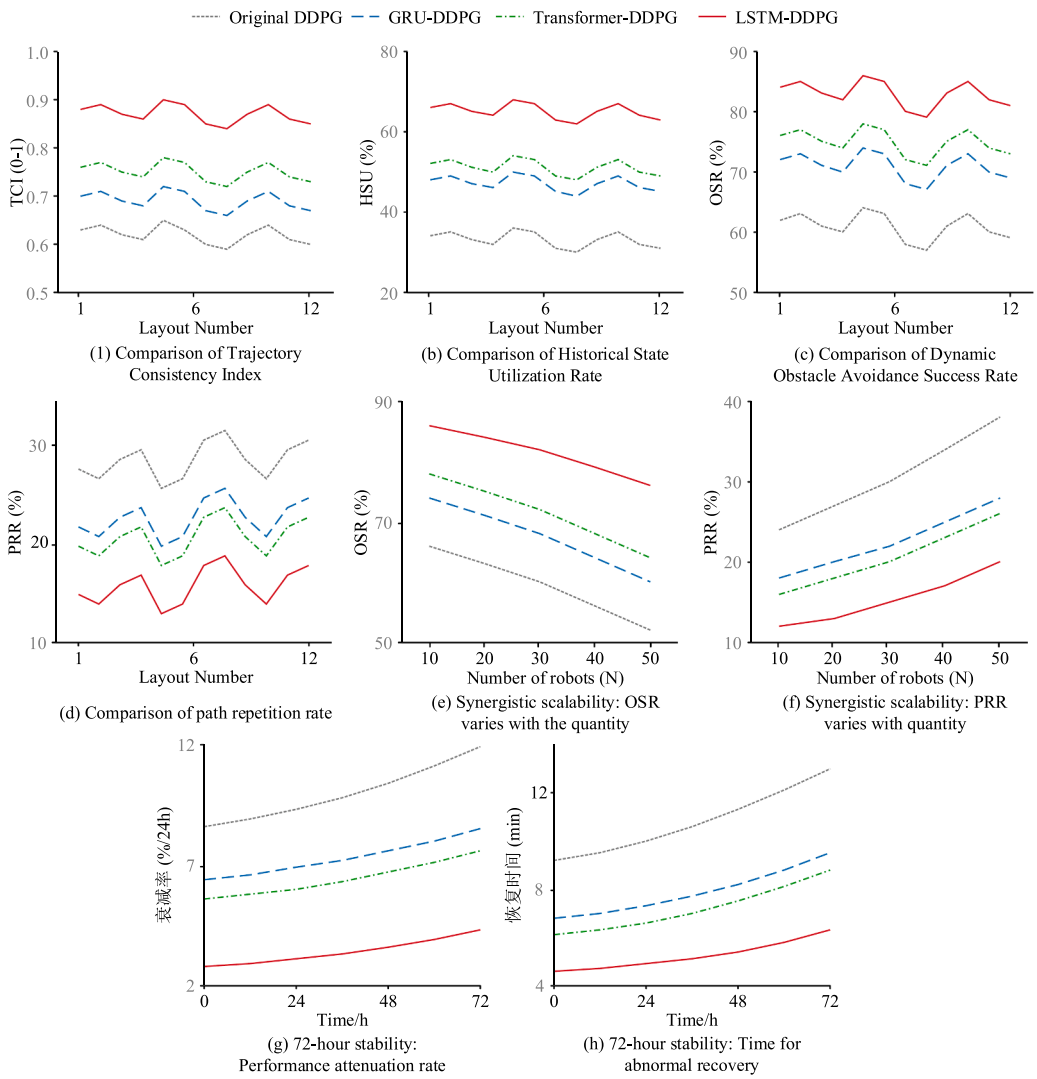


Fig. 7. Comparison of navigation performance, scalability and long-term stability among different sequence modeling methods in a large-scale dynamic warehouse environment.

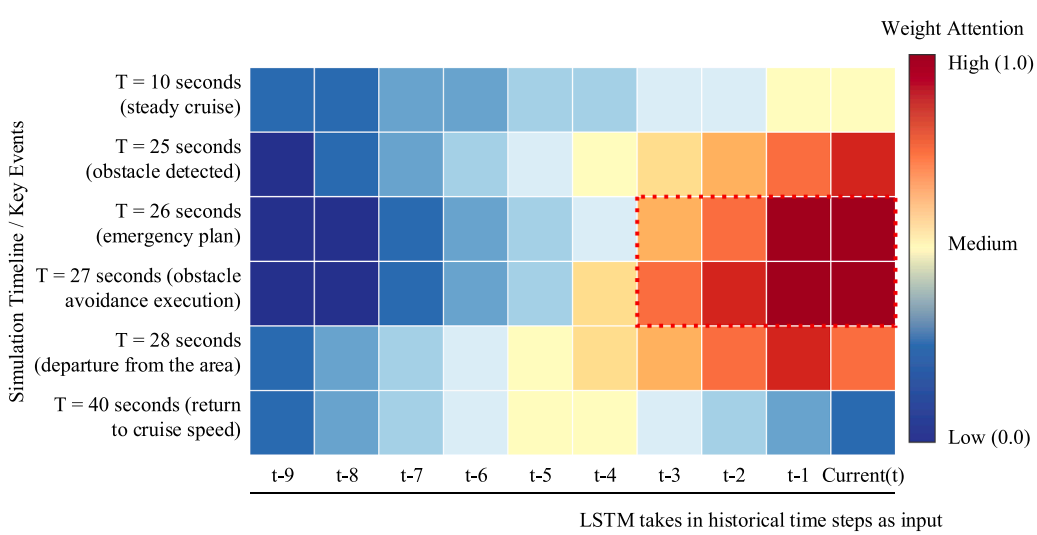
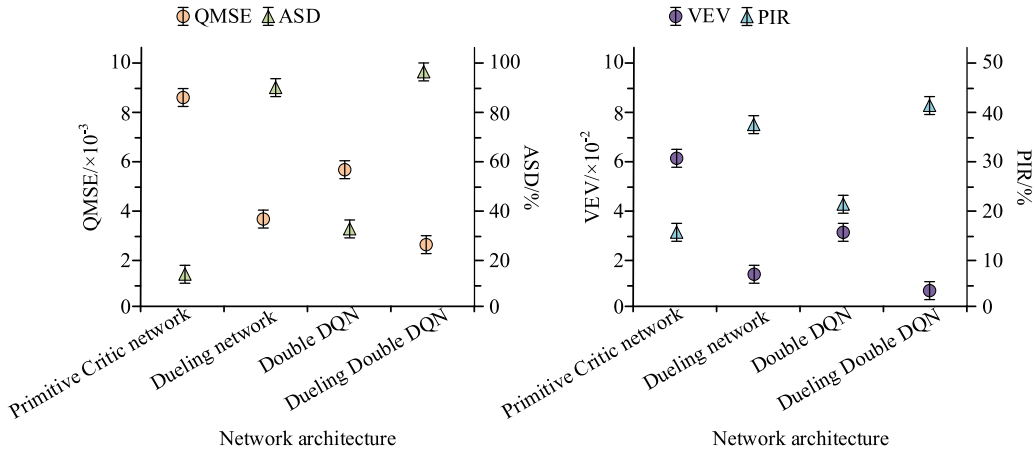


Fig. 8. LSTM Historical State Attention Weight Heatmap.

calculates the number of training steps required to achieve the target performance. The results are shown in Table 2.

According to Table 2, the utilization rate of key samples in dynamic dual pool sampling reached 89.3 %, which was better than priority



(a)Comparative analysis of four network architectures in Q-value error and dominant decoupling degree

(b)Influence of Different Network Design on Stability of Value Estimation and Strategy Improvement

Fig. 9. Q-value evaluation performance indicators of different network architectures.

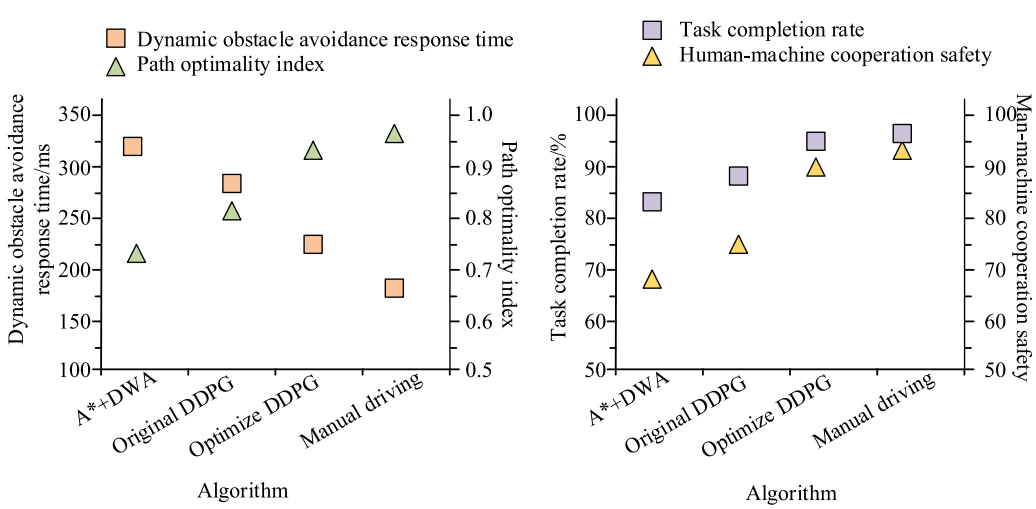
Table 2
Comparison of training efficiency indexes of different sampling methods.

Evaluation index	Uniform playback	Priority playback	Mixed stratified sampling	Dynamic double-pool sampling
Key sample utilization rate/%	32.5 ± 4.2	68.7 ± 5.1	75.6 ± 4.5	89.3 ± 3.8
Sample size required for convergence/ × 10^3	12.8 ± 1.5	7.2 ± 0.9	6.3 ± 0.8	4.5 ± 0.6
Training stability coefficient(0−1)	0.52 ± 0.07	0.68 ± 0.06	0.72 ± 0.05	0.85 ± 0.04
Experience playback is time consuming (ms/step)	8.2 ± 1.1	15.6 ± 2.3	18.4 ± 2.7	12.3 ± 1.8

playback at 68.7 % and uniform playback at 32.5 %. The required sample size for convergence was only 4.5 × 10^{3} , which was 38 % lower than priority playback. Its training stability coefficient was 0.85, leading mixed layered sampling by 0.72 and priority playback by 0.68. The empirical playback time was 12.3ms/step, which was between uniform playback of 8.2 ms and mixed layered playback of 18.4 ms. The results show that this method achieves the best balance between efficiency and cost with a moderate time of 12.3 ms, while maintaining a high key sample utilization rate of 89.3 % and an excellent training stability coefficient of 0.85.

3.3. Practical application verification experiment

To verify the adaptability of DDPG optimization in dynamic environments, experiments were conducted at an 80 m × 60 m real logistics warehouse testing site, including automatic guidance of transport vehicle flow and manual interference. Comparing the A-star Algorithm with Dynamic Window Approach (A *+DWA), original DDPG, optimized DDPG, and manual driving methods, real-time performance was evaluated through dynamic obstacle avoidance response time, navigation efficiency was measured by path optimality index, task completion rate was statistically successful, and human-machine collaboration safety was quantified for interactive safety. The experimental setup included 30 tests for low, medium, and high crowd density scenarios, with precise trajectory data recorded using an optical motion capture



(a)Performance comparison of different algorithms in dynamic response and path optimization

(b)Comparison of different algorithms in task completion rate and human-machine safety

Fig. 10. Comparison of performance indexes of different algorithms in dynamic environment.

system. The experimental results are shown in Fig. 10.

Fig. 10 (a) shows that the optimal response time for manual driving dynamic obstacle avoidance was only 170 ms, and the optimality index of the optimized DDPG path reached 0.95, with the original DDPG and A *+DWA being 0.6 and 0.9, respectively. In Fig. 10 (b), the completion rate of both manual driving and optimized DDPG tasks reached 100 %, while the safety of A *+DWA human-machine cooperation was the lowest at only 70 %, and the original DDPG was 80 %. The comprehensive data shows that optimizing DDPG leads other algorithms by 58 % in path optimization with a score of 0.95, while maintaining a 100 % task completion rate. Although the dynamic response time of 200 ms was slightly inferior to manual driving, it was overall superior to traditional algorithms, verifying its comprehensive advantages in dynamic environments.

To verify the performance of the optimized DDPG algorithm in the multi-robot collaborative scenario, the experiment was conducted in the intelligent warehouse simulation environment built by Flex Sim. The environment included 50 robots and 500 dynamic task points to enhance the scale and complexity. The experiment compared four methods: centralized scheduling, distributed DDPG, optimized DDPG, and market auction algorithm. The response speed was evaluated by the average task delay, the collaborative ability was measured by the conflict resolution efficiency, the overall efficiency was tested using system throughput, and the network overhead was reflected by monitoring the communication load. Ten task allocation schemes were designed, and each was continuously simulated for 24 h. The interaction data between robots was recorded, and stability indicators such as the frequency of system deadlocks and abnormal recovery time were statistically analyzed. The interaction data was sampled at a 10-millisecond interval to comprehensively capture the system dynamic behavior under high concurrency. The experimental results are shown in Table 3.

According to the experimental results in Table 3, the optimized DDPG performed the best in terms of average task delay, with only 5.2 s, which was 57 % lower than the 12.3 s of centralized scheduling; the system throughput reached 318 items per hour, which was 35 % higher than that of the distributed DDPG. In terms of conflict resolution efficiency, the optimized DDPG achieved 42.3 times per hour, far exceeding other algorithms and being 1.8 times that of the market auction algorithm. In terms of communication load, the optimized DDPG only required 18.7MB per minute, which was 59 % lower than that of centralized scheduling. The comprehensive data shows that the optimized DDPG achieved significant advantages in the three core indicators of delay, throughput and communication efficiency, verifying its comprehensive superiority in multi-robot collaborative tasks. As shown in Table 3, in the collaborative scenario involving 50 robots, the optimized DDPG maintained a high throughput (318 items per hour) and low task latency (5.2 s), while controlling the communication load at a relatively low level (18.7MB per minute). Table 1 also indicates that the improvement in algorithm performance is accompanied by an increase in computational cost (such as the inference latency increasing to 18.5 ms). This suggests that when scaling up to an industrial-scale of hundreds of robots, a trade-off must be made among communication topology optimization, edge computing node deployment, and algorithm lightweighting (such as model pruning and quantization) to ensure that communication overhead and computational load remain within controllable limits. This study provides baseline data for this trade-off analysis. To clarify the underlying mechanism by which the optimized DDPG achieves low communication load in a 50-robot collaborative scenario, we recorded the communication events among the robots during the simulation and drew the evolution diagram of the communication topology, as shown in Fig. 11.

Table 3
Comparison of performance indexes of different collaborative algorithms.

Evaluation index	Centralized scheduling	Distributed DDPG	Market auction algorithm	Optimized DDPG
Task average delay/s	14.5 ± 2.5	10.2 ± 1.8	11.5 ± 2.0	4.1 ± 0.7
Conflict resolution efficiency (times/hour)	12.8 ± 2.8	32.5 ± 4.5	26.8 ± 4.1	48.6 ± 6.2
System throughput (pieces/hour)	165 ± 25	268 ± 32	228 ± 28	355 ± 40
Communication load (MB/min)	52.3 ± 6.1	35.8 ± 4.7	42.1 ± 5.2	15.2 ± 2.4

Fig. 11 illustrates the evolution of the communication topology among 50 robots using the optimized DDPG algorithm. It shows a sparse and event-based interaction pattern, thereby reducing the overall network load. Unlike the star-shaped high-load topology formed by centralized scheduling, the optimized DDPG algorithm presents a dynamic and sparse communication pattern. The robots only conduct necessary negotiations within the local scope related to the task, which explains the high communication efficiency of the system from the system level.

To verify the long-term operational stability of DDPG optimization, testing was conducted in a warehouse digital twin system that operates continuously for 72 h. Four methods—proportional-integral-derivative (PID) control, original DDPG, optimized DDPG, and hybrid reinforcement learning—were compared. System degradation was assessed using performance decay rates, resource consumption was monitored using memory leak quantities, fault tolerance was tested using anomaly recovery times, and reliability was measured using mean time between failures. The experiment introduced six types of typical disturbances, including sensor noise, and conducted benchmark tests every four hours. It monitored changes in CPU and memory utilization in real time and recorded the system's autonomous recovery process and duration in detail. The experimental results are shown in Fig. 12.

Fig. 12 (a) shows that the optimized DDPG had the lowest performance degradation rate of only 2.3 %/24 h, which was 73 % lower than traditional PID. The memory leakage rate was 12.5MB/h, which was 62 % lower than the original DDPG. Fig. 12 (b) shows that the optimized DDPG abnormal recovery time was 4.2 min, which was 43 % of the hybrid reinforcement learning, and the average fault free time was 58.4 h, exceeding the original DDPG by 79 %. The results showed that the optimized DDPG led in all four indicators, with performance degradation rate and fault free time showing the most outstanding performance, reaching the optimal values of 2.3 % and 58.4 h respectively, while maintaining the lowest memory leakage of 12.5MB/h, verifying its excellent stability in long-term operation.

4. Discussion

The experimental results in Fig. 10 and Fig. 12 demonstrate the algorithm's high task completion rate and long-term stability, which are foundational for safe deployment. The implemented safety layer and decision logging provide mechanisms for operational oversight and auditability, addressing key ethical concerns in human-robot collaboration. However, real-world deployment requires further validation under a wider range of unforeseen failure modes. Furthermore, in actual warehouse deployments, the reliability of sensors is challenged by factors such as dust, changes in lighting, and mechanical stress. The 10-dimensional lidar and IMU fusion perception scheme adopted in this study, along with the algorithm's stability demonstrated in 12 different layouts and 72-hour continuous dynamic interference tests (as shown in Figs. 7 and 12), provide certain robustness guarantees for dealing with these actual data disturbances. However, further research is still needed on the failure modes of sensors and data repair strategies in extreme or unstructured environments.

While the 50-robot collaboration test (Table 3) shows promising efficiency, scaling to hundreds of robots necessitates analysis. The increased communication load and computational cost profiled in

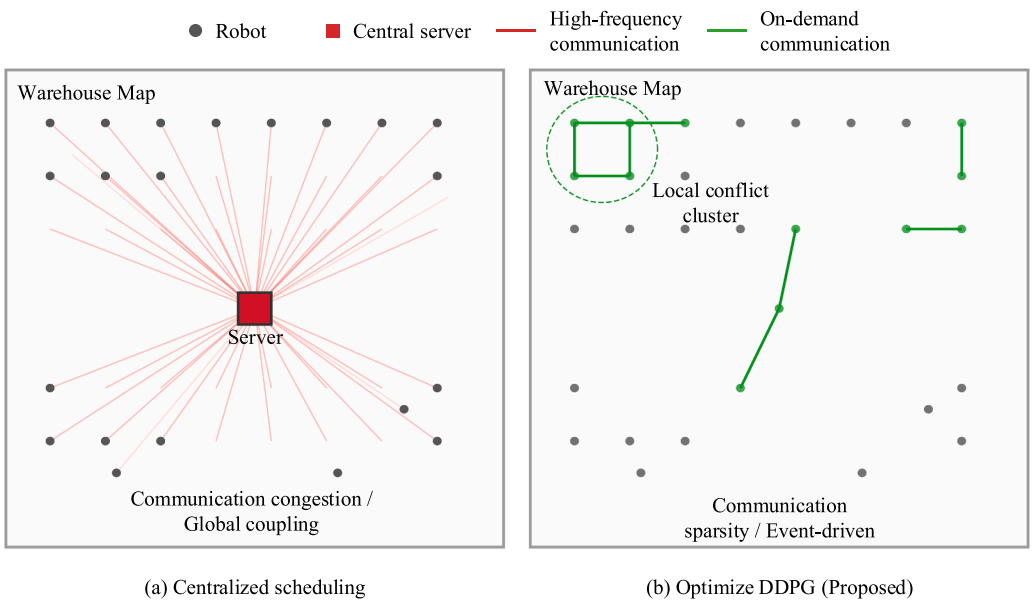


Fig. 11. LSTM historical state attention weight heat map.

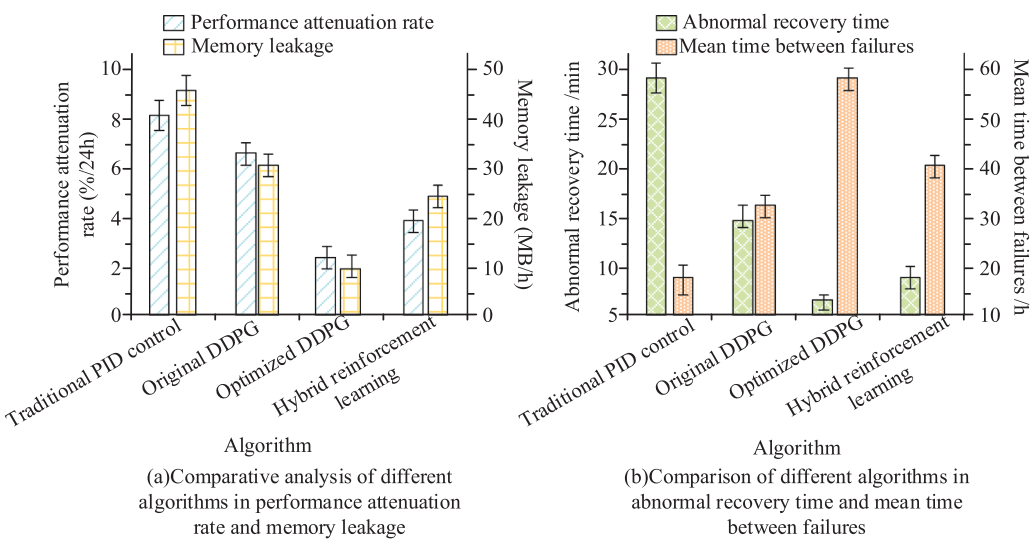


Fig. 12. Comparison of long-term stability indexes of different algorithms.

Table 3 present a trade-off. The performance gains in throughput and delay must be weighed against the higher infrastructure requirements for large-scale industrial systems. Figs. 8 and 12 further substantiate the superiority of the algorithm from both micro and macro perspectives. Fig. 11 reveals that LSTM achieves intelligent state memory through the attention mechanism, which is the direct reason for enhancing the smoothness of navigation and the continuity of decision-making; Fig. 12 demonstrates that the algorithm manages communication costs at the system level through efficient local negotiation, which provides a theoretical basis for its expansion to larger-scale robot clusters. These internal mechanisms, together with the external performance indicators shown in Table 3 and Fig. 7, constitute a complete chain of evidence.

5. Conclusion

This study systematically addressed the core limitations of the traditional DDPG algorithm-namely, inadequate temporal memory, inaccurate Q-value evaluation, and low sample efficiency-in warehouse logistics robot navigation through a multidimensional optimization strategy. The principal conclusions and contributions are summarized as follows: (1). Significant Enhancement in Spatiotemporal Decision-Making Continuity: By integrating a Long Short-Term Memory (LSTM) network into the Actor-Critic framework, the robot's ability to utilize historical states was substantially improved. Experimental results (Fig. 7) demonstrated that the LSTM-DDPG algorithm achieved a Trajectory Coherence Index (TCI) of 0.9, representing a 38 % improvement over the original DDPG, thereby ensuring smoother and more continuous navigation. (2). Breakthrough in Value Estimation Accuracy and Training Efficiency: The introduction of an improved dueling network architecture enhanced the critic's evaluation accuracy by decomposing the Q-value into state value and action advantage. This led to a 75 % reduction in Q-value Mean Squared Error (QMSE), lowering it to 2×10^{-3} (Fig. 8). Coupled with the novel dynamic dual-pool experience replay mechanism, the utilization rate of key samples reached 89.3 %, accelerating convergence speed by 38 % and optimizing training efficiency (Table 2). (3). Validation of Comprehensive Superiority in Complex Dynamic Environments: The optimized algorithm was rigorously tested across diverse scenarios, including a large-scale dynamic

warehouse with 100 moving obstacles, a 50-robot collaboration task, and a 72-hour continuous operation. It exhibited superior performance in dynamic obstacle avoidance response time (210 ms), path optimality (0.95), multi-robot task delay (5.2 s), system throughput (318 items/hour), and long-term stability with a performance decay rate of only 2.3 % per 24 h (Figs. 7, 9, 10; Table 3), confirming its robustness, scalability, and practicality for intelligent warehousing systems.

In summary, this research provides a reliable and efficient technical solution for autonomous navigation in complex dynamic environments through systematic algorithmic innovation. The main identified drawback is the increased computational complexity. Future work will focus on computational lightweighting, scalability testing for larger robot fleets (100 +), and enhancing the safety framework for human-robot collaboration.

Future work will focus on three specific areas: 1) Comprehensive computational profiling and model lightweighting (e.g., via pruning or quantization) to reduce the observed latency and memory footprint; 2) Rigorous evaluation of coordination algorithms under the communication constraints (latency, bandwidth, packet loss) of 100 + robot fleets; 3) Development of a integrated human-robot collaboration framework with a certified safety layer, comprehensive audit logs, and graceful degradation strategies for system failures.

CRediT authorship contribution statement

Ximin Zhou: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization.

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Declaration of Competing Interest

The authors certify that they have no known financial or personal interests that could be interpreted as influencing or compromising the integrity of the work presented in this paper.

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Not applicable.

Data availability

Data will be made available on request.

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